Applied Cryptography

6COSC019W- Cyber Security

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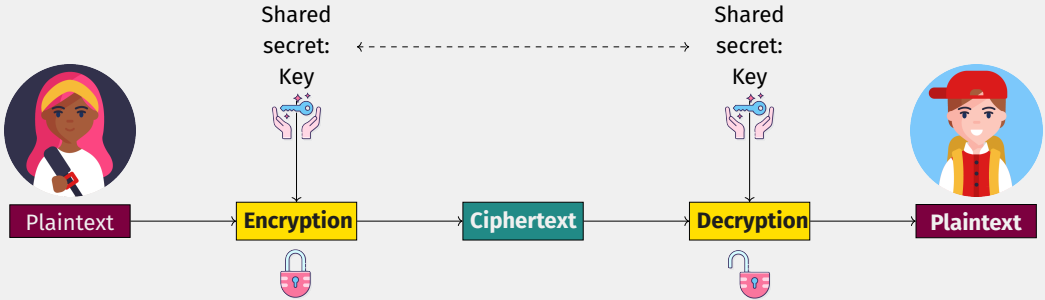
Overview

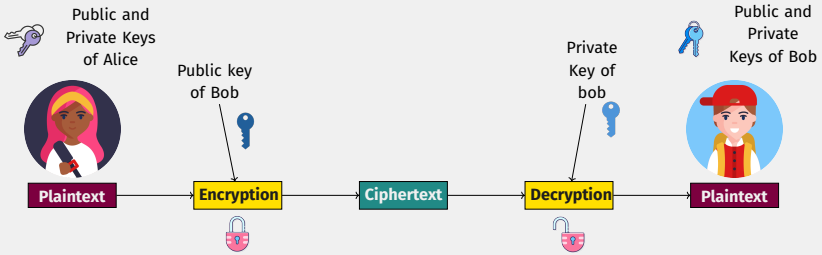
1. Stream and Cipher modes
2. Cryptographic algorithms
3. Keys Distribution
4. Hashing Algorithms
5. Digital signatures

Symmetric Encryption - A recap



Asymmetric Encryption

- Commonly known as public key cryptography
- Uses a pair of keys: a private key that is kept secret and a public key that can be sent to anyone
  - Security reliant upon resistance to deducing one key, given the other
- Public key systems, because of their design, also form the basis for digital signatures
  - A cryptographic method for securely identifying people

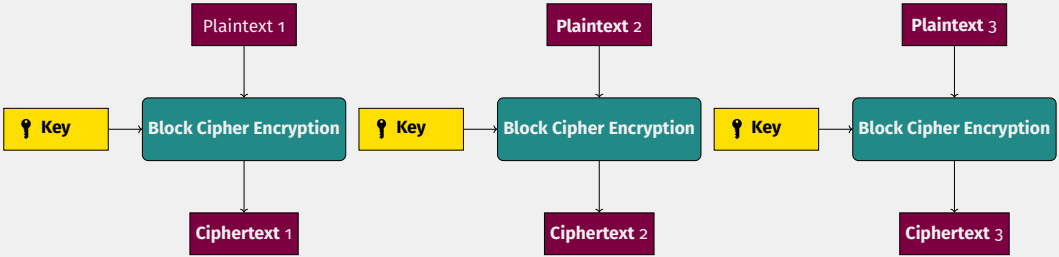


Cipher Modes

- When dealing with block algorithms, there is a need to deal with multiple blocks of identical data to prevent multiple blocks of cyphertext that would identify the blocks of identical input data
  - There are multiple modes of operation
    - Electronic Codebook (ECB)
    - Cipher Block Chaining (CBC)
    - Counter Mode (CTR)

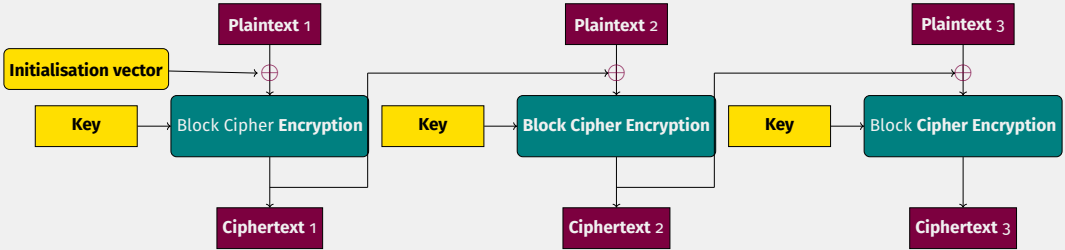
Electronic Code Book (ECB)

- A block cipher mode describes the way a block cipher encrypts and decrypts a sequence of message blocks.
- Electronic Code Book (ECB) Mode (is the simplest):
  - Block P[i] encrypted into ciphertext block C[i] = EK(P[i])
  - Block C[i] decrypted into plaintext block M[i] = DK(C[i])



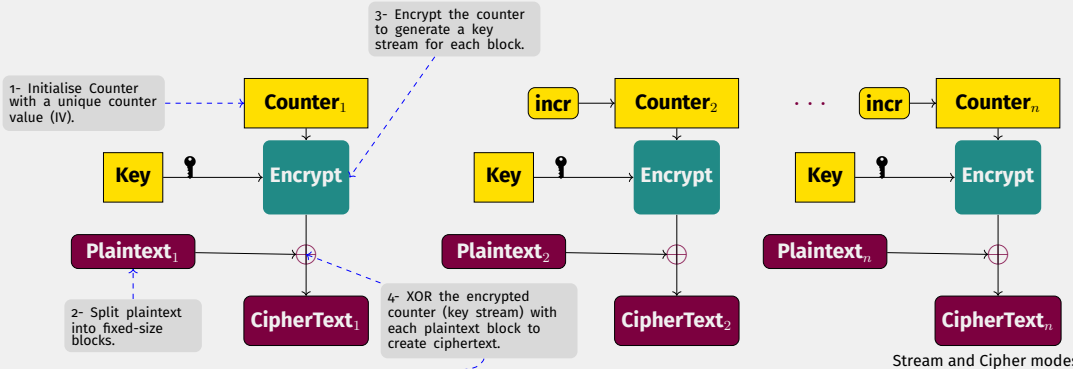
Cipher Block Chaining (CBC) Mode

- In Cipher Block Chaining (CBC) Mode
  - The previous ciphertext block is combined with the current plaintext block $C[i] = EK(C[i-1] \oplus P[i])$
  - $C[-1] = V$, a random block separately transmitted encrypted (known as the initialization vector)
  - Decryption: $P[i] = C[i-1] \oplus DK(C[i])$



Counter Mode (CTR)

- The Counter Mode or CTR is a simple counter-based block cipher that encrypts each counter value and use it as an input to XOR it with plaintext which results in ciphertext block.
  - This turns block cipher into a stream cipher.
- The CTR mode is independent of feedback use.

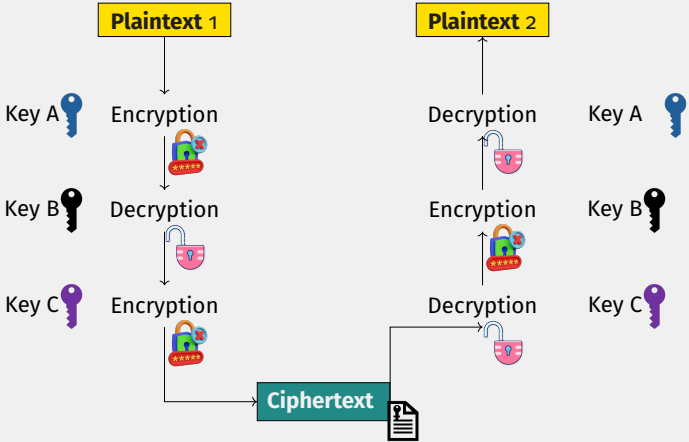


DES

- Data Encryption Standard (DES) is a block cipher
  - Block size of 64 bits—64 bits of plaintext outputs 64 bits of ciphertext
  - 56-bit key length
  - Performs a substitution and permutation based on the key 16 rounds on every 64-bit block
- Problems with DES
  - Weak key: less secure than the majority of keys allowed in the keyspace of the algorithm
  - Semi-weak keys: two keys that will encrypt plaintext to identical ciphertext
- DES was found to be reasonably secure
  - Has been for more than two decades

3DES

Nested application of DES with three different keys K_A, K_B, and K_C



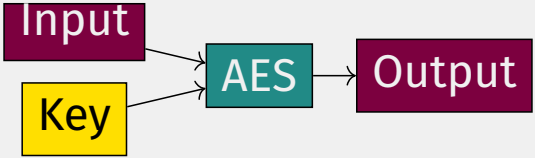
$$C = EK_C(DK_B(E_KA(P))); P = DK_A(E_KB(D_KC(C)))$$

3DES

- Depending on the variant, it uses either two or three keys
- Multiple encryption; goes through the DES algorithm three times
- The Encrypt and Decrypt combination is equivalent to DES when KA=KB=KC (backward compatible)
- Stronger than DES but has similar weakness
- The longer key length makes it more resistant to brute force attacks
- Still popular and widely supported
- AES has taken over as the symmetric encryption standard

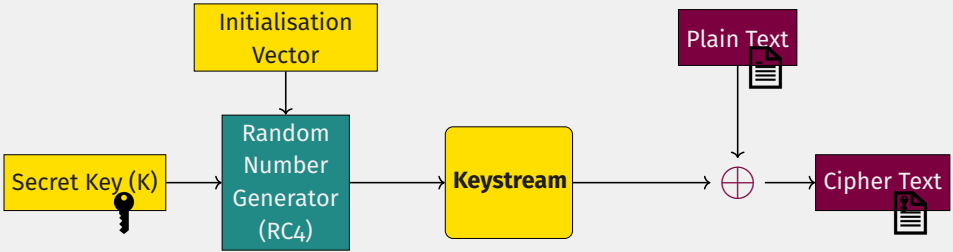
Advanced Encryption Standard (AES)

- A Cryptography algorithm that uses Symmetric encryption with block cipher of 128 bits.
- AES-256 is the symmetric encryption algorithm of choice nowadays. It won the NIST 2001 Competition.
- In **practice**, there has been no demonstrated attack that can break AES encryption. Even brute force operations is infeasible with current technologies $2^{128} = 3.4 * 10^{38}$.
- Variable key sizes used (128, 192, 256 bits)



Rivest Cipher 4 (RC4)

- Symmetric encryption algorithm designed in 1987 by Ron Rivest for RSA Security and remained a trade secret until 1994.
- keys length: up to 2,048 bits
- RC4 was popular for its remarkable speed and simplicity in software-based encryption in applications such early versions of TLS/SSL, WEP and WPA.
- It is now deprecated due to its several weaknesses.



RSA Algorithm

- One of the first public key cryptosystems invented
  - Published in 1977
  - Used for encryption and digital signatures
- Uses the product of two very large prime numbers
  - Works on the principle of difficulty in factoring such large numbers
- A simple algorithm that has withstood the test of more than 20 years of analysis
- Process divided into two parts:
  - Keys generation
  - Encryption/Decryption

RSA Keys generation

Choose two large primes p and q

Compute $n = p \cdot q$

$\phi(n)$ represents the number of integers relatively prime to (n), called the Euler quotient function

Compute $\phi(n) = (p-1)(q-1)$

usually e starts from 65537 as it provides better resistance against small exponent attacks compared to smaller values like

Choose e such that $1 < e < \phi(n)$ and $\gcd(e, \phi(n)) = 1$

Compute d where $e \cdot d \equiv 1 \mod \phi(n)$

The modular inverse of $e \mod \phi(n)$ is d to ensure that e and d are mathematically linked for encryption and decryption in RSA.

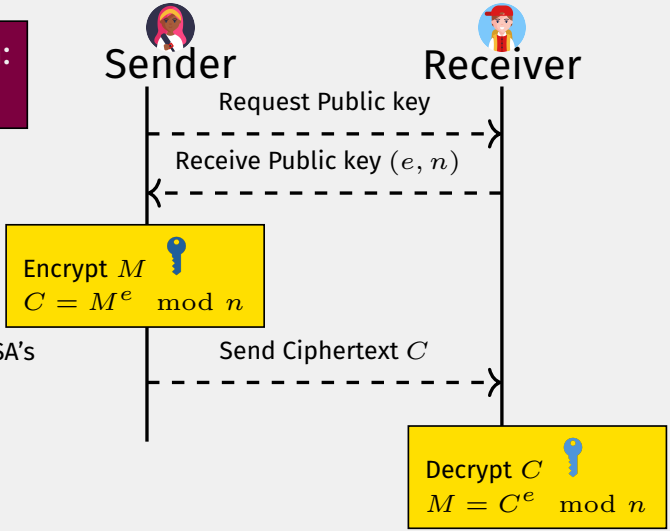
Public Key 🔑 $K_{pu} : (e, n)$

Private Key 🔑 $K_{pr} : (d, n)$

RSA Encryption/Decryption

Condition:
$M < n$

This condition ensures the mathematical validity and reversibility of RSA encryption and decryption. This constraint is a fundamental property of RSA's modular arithmetic



Key Distribution

- For symmetric encryption to work, the two parties to an exchange must share the same key, and that key must be protected from access by others
- Key distribution techniques
  - The means of delivering a key to two parties that wish to exchange data, without allowing others to see the key
  - **Paper distribution** : It requires no technology to use.
  - **Digital distribution**
    - can be in the form of digital storage or email but must be protected during transmission.
    - The Internet Security relies on this form of Keys distribution called public key infrastructure (PKI).
  - **Hardware distribution**
    - Keys Distributed via **hardware** such as a smart card, or a plug-in module.
    - The advantage is that no copies exist outside of these components.

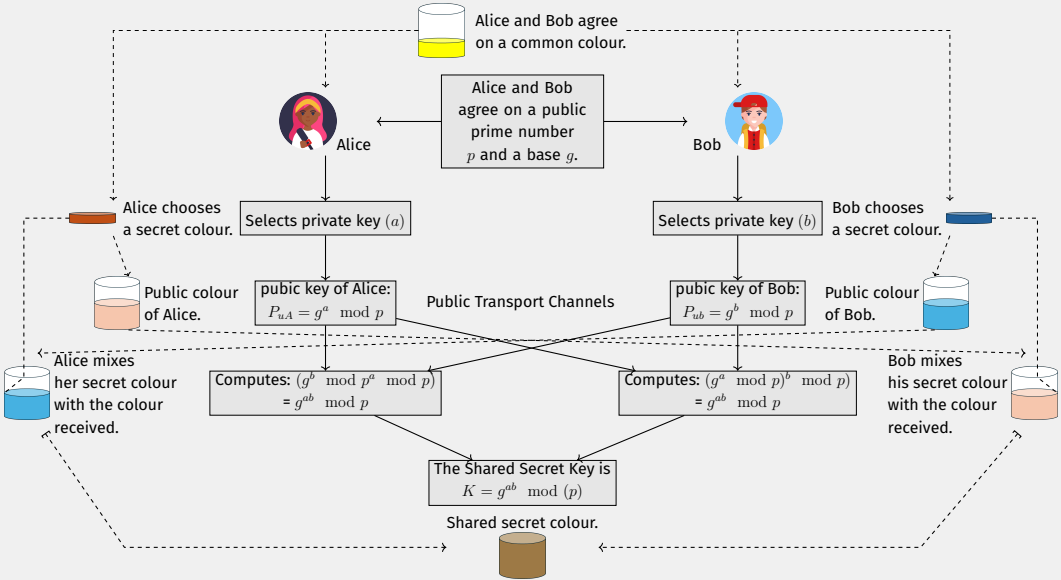
Session Keys

- Symmetric keys used for encrypting messages during a communication session
  - Generated from random seeds
  - Used for the duration of a communication session
- Session key advantages
  - Symmetric encryption, speed, strength, simplicity
  - Significant levels of automated security

Public Key Infrastructure (PKI)

- PKI provides a mechanism through which two parties can establish a trusted relationship even if the parties have no prior knowledge of one another
- PKI brings trust, integrity, and security to electronic transactions
- PKI framework used to manage, create, store, and distribute keys and digital certificates
- One of the most common PKI protocols in use today is the Diffie Hellman protocol
  - The purpose of the algorithm is to enable two users to exchange a secret key securely that can then be used for subsequent encryption of messages.
  - Plays a role in the electronic key exchange method of Transport Layer Security (TLS), Secure Shell (SSH), and IP Security (IPSec) protocols
  - The algorithm itself is limited to the exchange of the keys.

Diffie-Hellman Key exchange



Diffie-Hellman Key Exchange process

1. **Common colour**: Both Alice & Bob agree on using a common colour.
2. **Secret Colours**: Alice & Bob each choose a secret colour.
3. **Mixing Colours**: Alice & Bob mix their secret colours with the publicly agreed colours and value**exchange** them using the Public channel.
4. **The common colour**: Each respectively create the common secret using their secret colour.
5. Alice & bob now have a common secret colour (key) .

1. Alice & Bob agree on a public prime number and a base.
2. Alice & Bob chooses their secret values.
3. Alice & Bob Calculate their public values and then **exchange** them using the Public channel.
4. Each of them calculates the shared secret key using their private value and the other's public value.
5. Alice & bob now have a shared secret key.

Key Distribution Centres

- Rather than each organisation creating the infrastructure to manage its own keys, a number of hosts could agree to trust a **key-distribution center (KDC)**
- All parties must trust the KDC
- With a KDC, each entity requires only one secret key pair—between itself and the KDC
- Kerberos use the concept of a KDC.
  - Kerberos is one of the most commonly used KDCs .

Kerberos

- Key distribution and user authentication service developed at MIT
- Provides a centralised authentication server whose function is to authenticate users to servers and servers to users.
- Relies exclusively on symmetric encryption, making no use of public-key encryption

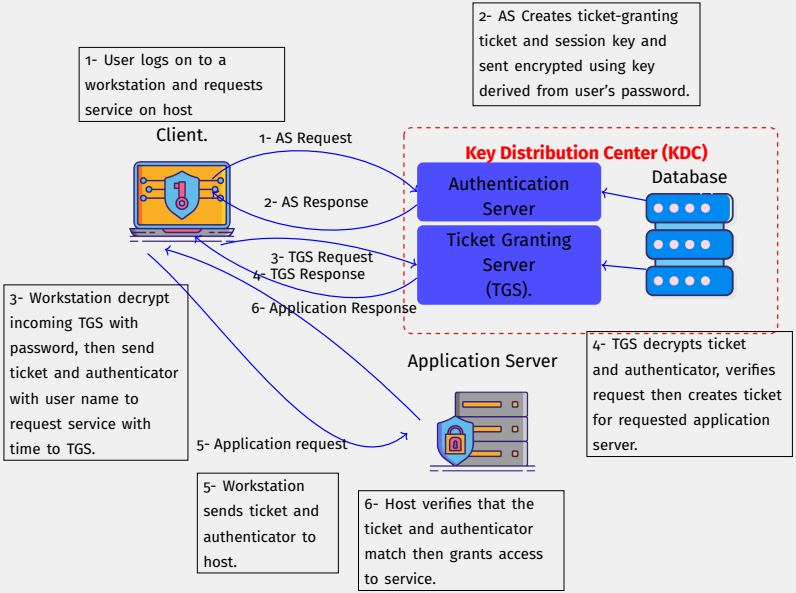
The problem Kerberos solves

- Assume an open distributed environment in which users at workstations wish to access services on servers distributed throughout the network. In this environment, a workstation cannot be trusted to identify its users correctly to network services.
- In particular, the following three threats exist:
  1. A user may gain access to a workstation and pretend to be another user operating from it.
  2. A user may alter the network address of a workstation so that the requests sent from the altered workstation appear to come from the impersonated workstation.
  3. A user may eavesdrop on exchanges and use a replay attack to gain entrance to a server or to disrupt operations.

How Kerberos works

1- User logs on to a workstation and requests service on host

2- AS Creates ticket-granting ticket and session key and sent encrypted using key derived from user's password.



3- Workstation decrypt incoming TGS with password, then send ticket and authenticator with user name to request service with time to TGS.

4- TGS decrypts ticket and authenticator, verifies request then creates ticket for requested application server.

5- Workstation sends ticket and authenticator to host.

6- Host verifies that the ticket and authenticator match then grants access to service.

Common Hashing Algorithms : Message Digest

- A message digest is a fixed-size numerical representation of a message.
  - Works in the same manner as SHA: secure method employed to compress the file and generate a computed output of a specified number of bits
  - A message digest must have the following properties
    - **Deterministic**: Same input $\rightarrow$ Same output.
    - **Irreversible**: Hard to reconstruct input from the digest.
    - **Collision-Resistant**: Hard to find two different inputs with the same digest.
- MD5 message digest algorithm—Takes an input of any arbitrary length and generates a 128-bit message digest that is computationally infeasible to match by finding another input.
- MD5 provides Integrity verification, only.

Common Hashing Algorithms: SHA256

- Secure Hashing Algorithm (SHA) variants are the most common variants of hashing functions found in commercial software
  - Secure Hash Algorithm (SHA-1)—Produces a 160-bit hash from a message of any arbitrary length.
  - SHA256 with a digest size of 256 bits is one the most widely used hash algorithms.

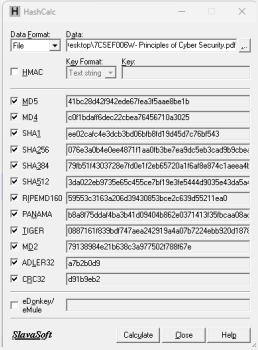
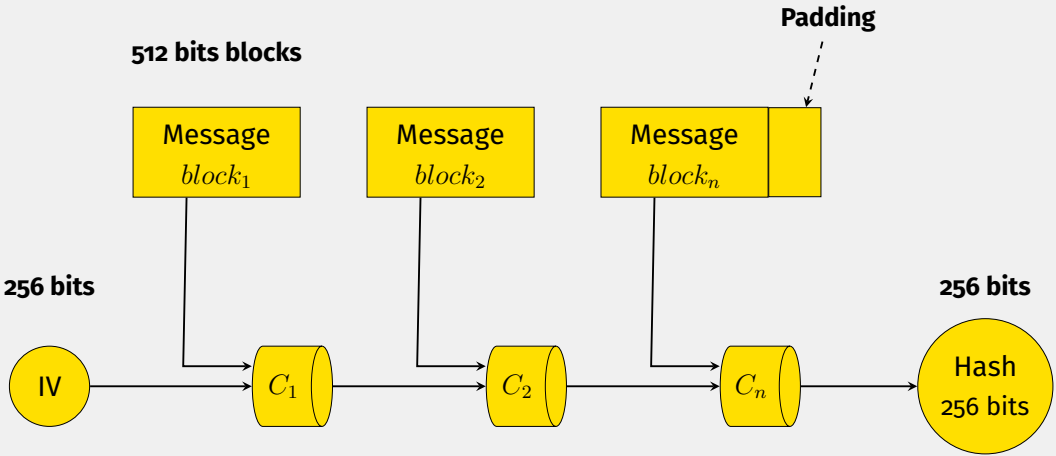


Figure 1: HashCalc -A program that accepts an input and produce a hash value to compare and verify with original file.

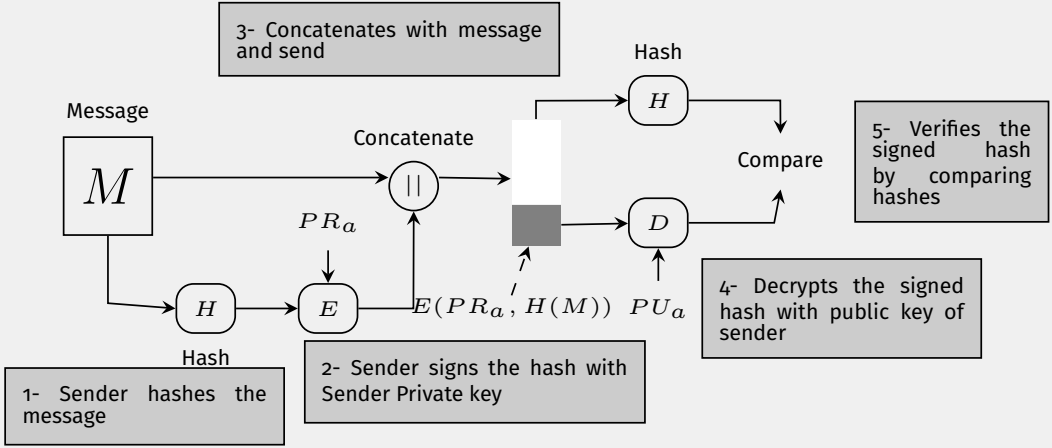
SHA256


Padding
512 bits blocks
Message
$block_1$
Message
$block_2$
Message
$block_n$
256 bits
256 bits
IV
C_1
C_2
C_n
Hash
256 bits

Digital Signatures

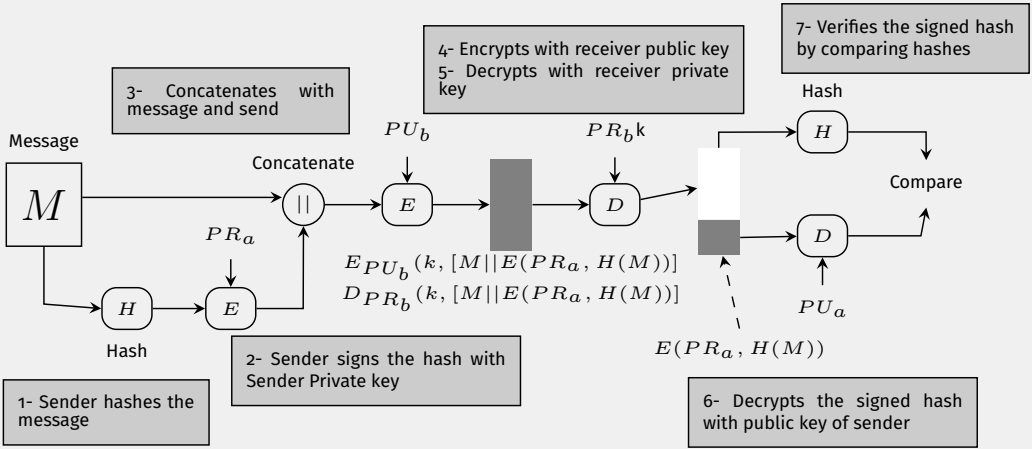
- A digital signature is a cryptographic implementation designed to demonstrate authenticity and identity associated with a message
- Based on hashing functions and asymmetric cryptography
  - Both encryption methods play an important role in signing digital documents
- It's operation is similar to that of the MAC
- The hash value of a message is encrypted with a user's private key
- Anyone who knows the user's public key can verify the integrity of the message
- An attacker who wishes to alter the message would need to know the user's private key
- Implications of digital signatures go beyond just message authentication

Digital Signature example without Confidentiality



This provides authentication. It also provides a digital signature because only the sender could have produced the encrypted hash code.

Digital Signature example with Confidentiality



This provides confidentiality in addition to authentication. It also provides a digital signature because only the sender could have produced the encrypted hash code.

Hashes vs Digital Signature

Method	Integrity	Authentication	Confidentiality	Non-Repudiation	Keys	Example
HMAC	✓	✓ (hash)	✗	✗	Symmetric	PGP (encrypted email)
CMAC	✓	✓ (symmetric key)	✗	✗	Symmetric	Hash + AES (file integrity)
Authenticated Encryption	✓	✓ (hash)	✓ (symmetric key)	✗	Symmetric	CHMAC + AES (API auth, TLS, file protection)
Digital Signature (no confidentiality)	✓	✗	✗	✓ (sender private key)	Asymmetric	Digital Signatures (RSA)
Digital Signature (with confidentiality)	✓	✓ (receiver public key)	✓ (receiver private key)	✓ (sender private key)	Asymmetric	Digital Signatures (RSA)

Table 1: Comparison of Hash-based authentication vs Digital Signatures

References

- The lecture notes and contents were compiled from my own notes and from various sources.
- **The lecture notes are very detailed. If you attend the lecture, you should be able to understand the topics.**
  - Chapter 2- Cryptographic Concepts, Hacker Techniques, Tools, and Incident Handling
  - Chapter 8- Encryption, Computer Security Fundamentals
  - Chapter 7- Cryptography, Fundamentals of Information Systems Security
- **Recommended Readings note:** Focus on what was covered in the class.